

PRIME-ANC: PATH-RATIO-INFORMED MODELING FOR EFFICIENT NEURAL FILTER SYNTHESIS IN ACTIVE NOISE CONTROL

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ABSTRACT

Changes in listener acoustics require active noise control (ANC) filters to be redesigned for new acoustic paths. We introduce PRIME-ANC, a shared neural synthesizer that learns a bounded, path-dependent log-magnitude correction to a regularized path-ratio base. Minimum-phase reconstruction and truncation produce finite-impulse-response (FIR) filters; training optimizes their noise-control performance. Across ten random training/test splits per dataset, PRIME-ANC achieves average held-out one-third-octave reductions of 18.81 and 17.76 dB over 50 Hz–5 kHz on a ten-path dataset and a public earphone database, respectively. On original earphone measurements, it improves upon the path-ratio base by 7.89 dB. Ablation studies support the contributions of both the analytic base and the path-dependent correction. Given calibrated paths for a held-out listener condition, PRIME-ANC generates a path-specific FIR without iterative optimization. With three Gauss–Newton updates, it reaches 21.43 dB reduction, approaching direct weighted least-squares design while producing lower amplification and root-mean-square control output.

Index Terms— active noise control, acoustic-path conditioning, amortized optimization, neural filter synthesis

1. INTRODUCTION

Listener anatomy and earphone fit alter acoustic paths, making rapid filter redesign important in feedforward active noise control (ANC) [1, 2]. Filtered-x least-mean-square (FxLMS) and normalized FxLMS (FxNLMS) adapt filters online using a secondary-path estimate [3]. Alternatively, acoustic-model-based design computes fixed filters for the target paths [4].

For rapid responses to changing noise, selective fixed-filter ANC (SFANC) selects predesigned filters [5], while

generative fixed-filter ANC (GFANC) combines subfilters derived from a broadband filter [6]. These filter resources remain system-dependent: GFANC can reuse its trained network with subfilters matched to the new system [7]. End-to-end control-filter generation (E2E-CFG) predicts complete filters from noise, but learns this mapping under fixed paths and generally requires retraining for a new acoustic environment [8]. Thus, transferring these noise-conditioned methods to a new acoustic system still entails preparing suitable filter resources or retraining the network.

Meta-learning reuses experience across acoustic conditions through learned initial control filters [9] and joint control-filter/secondary-path initializations [10]. Regional initialization also improves attenuation at unseen source positions before adaptation [11]. Primary and secondary paths determine how disturbance and control signals combine at the error microphone. We therefore combine cross-path learning with calibrated path information in a shared generator trained on sparse support paths, directly synthesizing a distinct causal finite-impulse-response (FIR) filter for each query without retraining or query-specific optimization.

PRIME-ANC realizes this mapping by combining a regularized path-ratio magnitude base with bounded, path-dependent log-magnitude corrections. Nonminimum-phase secondary paths or insufficient primary-path delay can prevent stable causal inversion [12]. Training therefore evaluates noise reduction, amplification and gain after minimum-phase reconstruction and truncation; paired interpolation extends sparse support coverage. The resulting FIRs also provide initializations for common Gauss–Newton updates. Experiments on two datasets assess held-path synthesis, the contributions of the base and conditioning, and the quality–design-cost trade-off under refinement.

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2. PRIME-ANC FILTER SYNTHESIS

2.1. Acoustic paths and the filter-construction task

Generator G_θ is trained on a *support set* $\mathcal{S}$ of path pairs. Each held-out *query pair* $q \notin \mathcal{S}$ requires a new filter. Let $p[n], s[n]$ denote primary and secondary impulse responses and $w[n]$ a real, causal FIR. Reference noise $x[n]$ gives disturbance $d[n]$ and residual $e[n]$,

$$d = p * x, \quad e = d - s * w * x, \quad (1)$$

where $*$ denotes convolution. Calibration measures residuals e_0, e_g under probes $W_0 = 0$ and $W_1 = g$, a known nonzero gain. Hann-windowed transfer estimates $\hat{H}_{yx}$ divide pooled cross-spectra by regularized reference autospectra and give [13]

$$\hat{P} = \hat{H}_{e_0 x}, \quad \hat{S} = (\hat{H}_{e_0 x} - \hat{H}_{e_g x})/g. \quad (2)$$

Synthesis and evaluation are

$$\begin{aligned} w_q &= G_\theta(\hat{P}_q, \hat{S}_q), \\ e_q &= (p_q - s_q * w_q) * x. \end{aligned} \quad (3)$$

One network pass and deterministic realization produce w_q without optimization. Estimates guide synthesis and refinement; true support paths supply supervision and true query paths supply residuals.

2.2. Path-ratio base and path-conditioned magnitude residual

The ideal target response is $W_{\text{ideal}}(f) = P(f)/S(f)$. We regularize its calibrated estimate as

$$W_{\text{reg}}(f) = \frac{\hat{P}(f)\hat{S}^*(f)}{|\hat{S}(f)|^2 + \epsilon_S^2}, \quad \epsilon_S = 10^{-4} \max_f |\hat{S}(f)|. \quad (4)$$

Regularization limits inversion near secondary-path spectral nulls [14]. Its magnitude $A_{\text{base}}(f) = |W_{\text{reg}}(f)|$ is the analytic base. Realization with zero learned correction yields the *ratio-base FIR*.

A shared convolutional neural network (CNN) predicts g_θ from calibrated path features, giving a bounded log-magnitude correction (Fig. 1):

$$\Delta_\theta = \frac{12 \ln 10}{20} \tanh g_\theta, \quad \log A_\theta = \log A_{\text{base}} + \Delta_\theta. \quad (5)$$

log is natural; corrections stay within ± 12 dB before realization.

Real-cepstral minimum-phase (MP) reconstruction supplies phase [15]; an inverse Fourier transform retaining samples 0–2047 yields the real, causal FIR. Input phase guides magnitude redistribution; reconstruction supplies

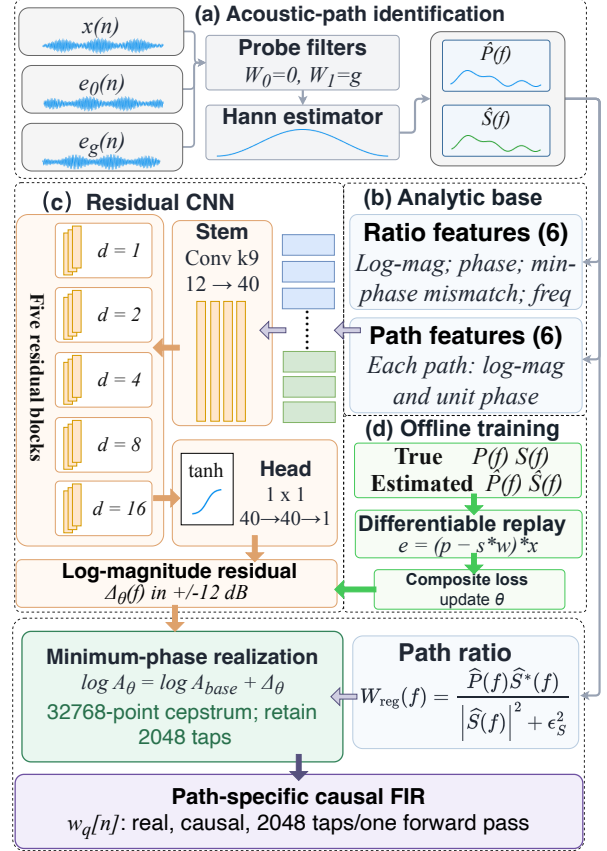


Fig. 1. PRIME-ANC: path-conditioned FIR synthesis from the magnitude base $A_{\text{base}} = |W_{\text{reg}}|$ and a shared neural generator.

output phase. Truncation can alter magnitude and MP structure, so loss is evaluated on the realized FIR.

For conditioning, MP reconstruction of A_{base} supplies unit phase u_{MP} ; $u = W_{\text{reg}}/|W_{\text{reg}}|$ is the ratio's unit phase. Six ratio channels encode log magnitude, normalized frequency, and real/imaginary parts of u and the phase mismatch uu_{MP}^* . Six path channels encode log magnitudes and unit phases. Only log magnitudes are standardized across frequency per path. The 151,401-parameter CNN uses group normalization and sigmoid linear unit (SiLU) activations.

2.3. Composite objective and support training

Training uses 3-s noise clips and differentiable path convolution. Let D, E be the Hann-windowed short-time Fourier transforms (STFTs) of disturbance and residual. Reduction in one-third-octave band b is $R_b = 10 \log_{10}[(\sum_{\mathcal{I}_b} |D|^2 + \epsilon)/(\sum_{\mathcal{I}_b} |E|^2 + \epsilon)]$, where $\mathcal{I}_b$ contains its STFT cells. Target-band noise reduction (NR) is $|\mathcal{B}|^{-1} \sum_{b \in \mathcal{B}} R_b$ for band centers $\mathcal{B}$ from 50 Hz to 5 kHz. Negative R_b indicates noise amplification (AMP),

measured as $[-R_b]_+$. We penalize its worst 10% of bands from 50 Hz to 8 kHz using conditional value-at-risk, $\text{CVaR}_{0.1}$. The objective is

$$\mathcal{L} = -\bar{R}_{50:5k} + 0.25 \text{CVaR}_{0.1}([-R_b]_+) + 0.02 \mathcal{L}_{24\text{dB}} + 10^{-3} \overline{\Delta^2} + 10^{-3} \mathcal{L}_{\text{smooth}}, \quad (6)$$

Here $\mathcal{L}_{24\text{dB}} = \overline{[20 \log_{10} |W| - 24]_+^2}$ averages over the realized response’s full Fourier grid; $\overline{\Delta^2}$ penalizes log-magnitude correction, and $\mathcal{L}_{\text{smooth}}$ is the mean squared second difference of realized taps. AdamW uses 6400 updates, batch size 2, learning rate 5×10^{-5} , and weight decay 10^{-4} ; the final checkpoint is evaluated.

To extend sparse training coverage, we optionally interpolate adjacent support path pairs i, j using one coefficient for both paths,

$$\tilde{p} = (1 - \alpha)p_i + \alpha p_j, \quad \tilde{s} = (1 - \alpha)s_i + \alpha s_j. \quad (7)$$

Estimated inputs and true training paths share α ; held-out conditions are unchanged.

3. EXPERIMENTAL SETUP

3.1. Datasets, training splits, and evaluation

All experiments use 48 kHz and 2048-tap filters. Ten-path comprises ten primary/secondary path pairs from Track 2 (ANC) of the 2026 CCF Advanced Audio Technology Competition [16]. The public PANDAR earphone acoustic-path database covers both ears of 23 participants using the same earphone platform [17]. We evaluate each measured secondary path and three gain/delay and spectral/tail variants, with its primary path unchanged. Both datasets use eight recordings from the Diverse Environments Multi-channel Acoustic Noise Database (DEMAND) [18], plus white and pink noise, at root-mean-square (RMS) level 0.05. Primary/secondary estimates have 2048/2048 taps on ten-path and 8192/2048 taps on PANDAR. PANDAR calibration simulates probe responses from the available paths and eight environmental recordings, with fixed paths across states and $g = 0.35$. Hann transfer estimates are pooled at six offsets. Supervised clips are drawn from 0–4 s; evaluation scores 7–10 s after 0.5-s warm-up.

Ten random identity-based splits select 3/10 paths or 7/23 participants as support, holding out 7 paths or 16 participants. Ears and variants stay with their participant. Each dataset/split has separately trained networks; fitting and interpolation use support pairs only. Each split adds 96 paired support mixtures, sampled with probability 0.75. Ten-path interpolates adjacent supports ordered by their first principal component; PANDAR interpolates adjacent support participants within each ear and secondary-path

condition. PRIME-ANC uses paired interpolation by default; its PANDAR training averages 161.6 s per model.

Scores average six 0.5-s blocks: NR over 50 Hz–5 kHz and AMP over 50 Hz–8 kHz, clipping $[-R_b]_+$ before averaging. Thus 5–8 kHz contributes to AMP only. Output RMS measures $w * x$. We average noises, ear/variant conditions and three seeds within each held path or participant, then held units within each split. Means and sample standard deviations (SDs) describe ten paired, equally weighted split means; the overlapping splits are not independent trials.

3.2. Direct synthesis and iterative references

Two analytic FIRs provide direct-construction baselines: ratio-base uses PRIME-ANC’s realization with zero correction; projected ratio retains inverse-transform samples 0–2047 of complex W_{reg} , without shifting. Both use 8192/32768-point Fourier grids on ten-path/PANDAR. First-order model-agnostic meta-learning (FOMAML) [9, 19] learns one initial FIR from true support paths, using 800 pooled Adam and 400 meta updates. Table 1 reports the same pre-adaptation FIR for all queries, without inner-loop updates.

Weighted minimum mean-square error (WMMSE) design supplies shared and per-query filters by minimizing

$$\min_{w \in \mathbb{R}^{2048}} \sum_{a \in \mathcal{A}} \sum_k \omega_{a,k} |\hat{P}_{a,k} - \hat{S}_{a,k} f_k^\top w|^2 + \lambda \|w\|_2^2. \quad (8)$$

Here $f_k^\top$ is a 2048-tap Fourier row on a 32768-point grid. Within each target band, reference-power weights are divided by the band’s Fourier-bin count, then normalized globally to unit mean over active bins. $\mathcal{A} = \mathcal{S}$ fits the shared filter; $\mathcal{A} = \{q\}$ fits query q , with $\lambda = 10^{-4} \sum_{a,k} \omega_{a,k} |\hat{S}_{a,k}|^2$. Shared filters use a direct Toeplitz solve. PANDAR query filters use a double-precision CPU Toeplitz solve or single-precision GPU preconditioned conjugate gradients (PCG) with fast Fourier transform (FFT) products. PCG uses zero initialization, diagonal preconditioning, relative residual tolerance 10^{-6} and budgets of 32–2048 iterations.

We adapt E2E-CFG’s convolutional–Transformer backbone and direct filter-generation approach [8] into a path-conditioned direct-tap baseline. Eight calibrated-path channels replace noise frames as inputs for predicting 2048 FIR taps. The head receives means and SDs of $\log |\hat{P}|$, $\log |\hat{S}|$ and $\log |W_{\text{reg}}|$ to retain amplitude scale. Training matches PRIME-ANC’s physical loss weights, 6400 updates, batch size 2, seeds and support-only paired interpolation on both datasets, using AdamW at 5×10^{-4} without the log-magnitude correction penalty. The final checkpoint is evaluated, and we do not interpret this result as a direct reproduction comparison with the original E2E-CFG setting.

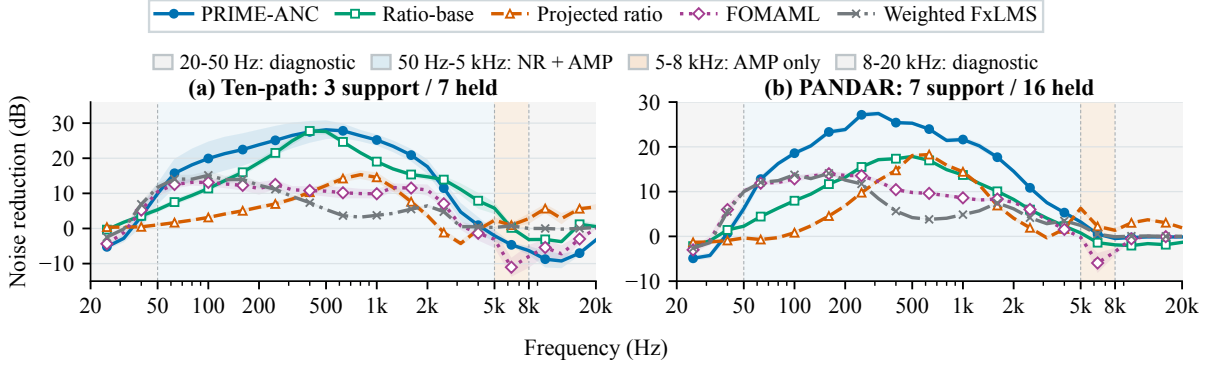


Fig. 2. Frequency-resolved NR (mean ± 1 split SD). Gray: 40-s weighted FxLMS per path/noise (PANDAR: original measurements).

Iterative FIR applies 8192 Adam steps (5×10^{-4}) per query from the ratio base to the reference-spectrum version of Eq. (6), complementing WMMSE’s quadratic criterion (Table 3). Single-precision Gauss–Newton (GN) refines PRIME, both analytic FIRs and FOMAML, plus direct WMMSE on PANDAR. With generator parameters fixed, it uses query estimates to minimize a tap-refinement objective $\mathcal{J}$ combining mean log third-octave residual and smooth worst-band risk (weight 0.05, temperature 1) on a 32768-point grid. Damping is $0.002 \max(\bar{h}, 10^{-6}) + 10^{-6}$, where $\bar{h}$ is the mean unregularized normal diagonal. PCG allows 16 iterations with relative residual tolerance 10^{-6} . Response-step RMS is capped at $10^{1/20} - 1$ times the query ratio-base RMS over active bins. Backtracking over 1, 1/2, 1/4 stops GN if no decrease reaches $10^{-5} \max(|\mathcal{J}|, 1)$. All starts share tap-update budget K ; $K = 0$ is unrefined. Updates use FFT operators, fused PCG and CUDA Graph replay. FOMAML, WMMSE, iterative FIR and GN share a reference power spectrum estimated from the ten noises over 0–4 s.

FxLMS and FxNLMS use true secondary paths and 50 Hz–5 kHz gradient weighting, with the common scoring protocol. Figure 2 shows weighted FxLMS on both datasets (PANDAR: original measurements). For Table 3, each path’s filter is adapted on white noise and frozen for evaluation on ten noises. Both share a validation-based stopping rule with a 240-s adaptation cap.

After runtime initialization, PANDAR single-query calls are timed three times on an RTX 5090, except for direct WMMSE on one Threadripper PRO 7975WX CPU thread. Condition medians are averaged within participants, seeds and splits as for NR. Core time covers query-specific filter construction and optimization from resident features and Fourier paths, including CPU-to-GPU transfer for WMMSE plus GN. Calibration, file access, model loading and runtime initialization are excluded.

Table 1. No query updates: NR (dB), mean $\pm$ SD over ten splits.

Method	Ten-path 3/7	PANDAR 7/16
PRIME-ANC, no interpolation	16.72 ± 2.08	16.22 ± 0.66
PRIME-ANC, paired interpolation	18.81 ± 2.57	17.76 ± 0.30
Ratio-base FIR (zero residual)	15.95 ± 0.29	10.09 ± 0.22
Projected complex-ratio FIR	5.95 ± 0.63	7.09 ± 0.28
FOMAML, pre-adaptation	9.43 ± 0.65	9.35 ± 0.22
Support-shared WMMSE	7.07 ± 2.46	8.81 ± 0.27
E2E-CFG (path-adapted) [†]	5.55 ± 4.43	9.75 ± 0.42

[†]Path-conditioned adaptation of E2E-CFG; see Section 3.2.

4. RESULTS AND DISCUSSION

4.1. Direct synthesis on held-out acoustic conditions

Direct synthesis gains 2.857/7.663 dB NR over the ratio base on ten-path/PANDAR (Table 1), with positive seed-averaged gains in 9/10 and 10/10 splits. Gains span much of the lower and middle target bands (Fig. 2); mean AMP is 0.770/0.134 dB. On original PANDAR measurements, PRIME improves every split, reaching 18.576 dB NR versus 10.686 dB for the ratio base, and 16.832 dB without interpolation. Modified conditions gain 7.588 dB with interpolation. Paired interpolation improves every split, adding 2.083/1.539 dB on ten-path/PANDAR (Table 1). Ten-path support/held ratios of 2/8–9/1 yield 16.28–22.76 dB NR; 3/7 uses sparse support.

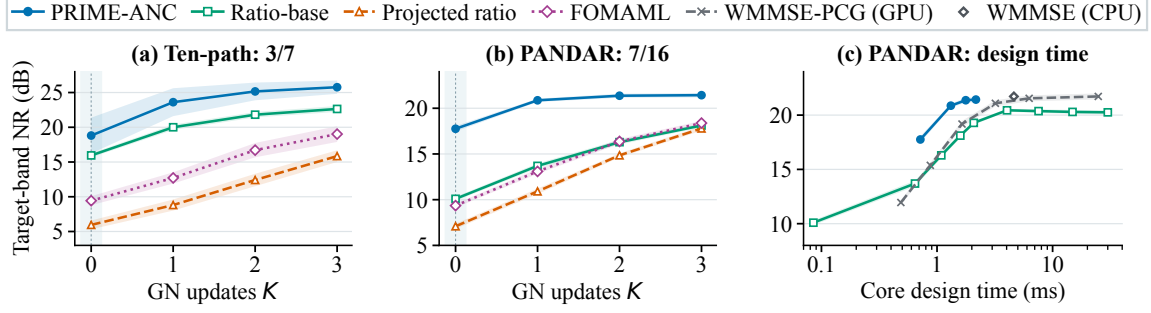


Fig. 3. Common GN updates (a,b) and PANDAR core design time (c). Shading: ± 1 split SD.

4.2. What the ratio base and conditioning contribute

To isolate the base connection, four output controls (Table 2) append summary statistics of $\log |\hat{P}|$, $\log |\hat{S}|$ and $\log A_{\text{base}}$ after the CNN trunk. They use $\log A = b \log A_{\text{base}} + (12 \ln 10/20)g_{\theta}$, with $b = 1$ or 0 and unbounded g_{θ} . Output phase is MP or MP plus $\pi \tanh v_{\theta}$ before the same FIR realization (151,641/151,682 parameters). Within each phase construction, only the base connection changes; features, initialization, augmentation, updates and seeds are matched. All four replace the correction penalty with $(\log |W| - \log A_{\text{base}})^2$; standard PRIME keeps the bound and penalty in Eqs. (5)–(6). We retain bounded PRIME across datasets, as the unbounded variant improves PANDAR but degrades ten-path NR.

Table 2. PANDAR construction and conditioning: NR/AMP (dB).

Filter construction	NR	AMP
PRIME-ANC (standard)	17.755	0.134
<i>Output construction controls</i>		
Ratio-base skip, MP	18.577	0.093
Ratio-base skip, MP + phase	18.616	0.093
No skip, MP	14.646	0.093
No skip, MP + phase	14.655	0.093
<i>Standard-model conditioning</i>		
Shuffled input phase	12.593	0.289
Path-independent correction	12.509	0.205

The base adds 3.93/3.96 dB with MP/phase-corrected outputs in all ten splits at similar AMP; phase correction adds < 0.04 dB. Path-independent correction and phase shuffling lose 5.246/5.162 dB NR, supporting path-dependent correction and input phase use.

4.3. Refinement and computational efficiency

Common GN tests initialization under identical query estimates and objective. PRIME leads analytic starts and FOMAML at all tested budgets ($K = 0$ –3) on both

Table 3. PANDAR quality and core design time (NR/AMP in dB). WMMSE + GN starts from the CPU direct solution.

Method	NR	AMP	RMS	Time (ms)
PRIME, direct	17.755	0.134	0.302	0.717
PRIME + GN(1)	20.869	0.136	0.303	1.317
PRIME + GN(2)	21.370	0.149	0.306	1.772
PRIME + GN(3)	21.430	0.156	0.307	2.169
Ratio-base + GN(4)	19.286	0.412	0.310	2.079
WMMSE, CPU direct	21.722	0.351	0.502	4.623
WMMSE + GN(3)	23.264	0.357	0.506	6.120
Iterative FIR	21.992	0.165	0.449	12430.125
Adaptive reference	NR	AMP	RMS	
<i>40–120 s white-noise adaptation</i>				
FxLMS	19.497	0.125	0.150	
FxNLMS	19.833	0.148	0.152	

datasets (Fig. 3a,b). At comparable core times of about 2.1 ms, three GN updates from PRIME achieve 2.14 dB higher NR than four GN updates from the ratio base (Fig. 3c). PRIME + GN(3) approaches direct WMMSE in NR with lower core time, amplification, and control RMS (Table 3). WMMSE + GN raises NR at higher cost.

5. CONCLUSION

PRIME-ANC combines an analytic path-ratio base with learned path-dependent log-magnitude corrections to synthesize causal FIRs from calibrated paths. Experiments on two datasets show gains over the ratio-base FIR on held-out paths and listener conditions, while ablations support both the base and path conditioning. On PANDAR, three GN updates approach direct WMMSE in NR with lower measured core design time, amplification, and control RMS. These results highlight PRIME-ANC as a promising dynamic framework for earphone ANC, en-

abling rapid, listener-specific filter generation with lower design overhead and robust noise control performance.

6. ACKNOWLEDGEMENT

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